

Speed Spiral

What drives technology to develop faster and more powerful processors?

They said it couldn't be done, but we're doing it. For years, I've heard industry observers say we're running out of time before today's technology impedes our progress. Technological change and progress is a bit like the chicken and the egg situation. Does the industry create the need or satisfy it? Before Sony developed the 'Walkman', no customer around the world had articulated the need for portable music. Customers didn't see the need for mobile phones either. Successful technologists and marketers around the world are futurists of

simple tasks like word processing, Internet surfing and e-mailing. But once we reach a surplus on computing power, we will have to figure out how we can actually utilise this big pool of excess resources.

On the horizon, a new influence is set to have an impact on the direction and the rate at which technology escalates. From the software spiral of the past and the present, a new generation of digital devices will provide some heavy guidance in the work that lies ahead.

The 'digital spiral' is an offshoot of what I'd

Digital devices, not software, will herald the next change...

sorts. They forecast and predict needs and the ones who do well consistently emerge as leaders.

Ever so often when we deliver faster processors or launch a new generation of devices, it meets with the pessimist's refrain of, "Who needs one?". The simple answer is "The consumer". The PC does a lot of things in our life now, from being an answering machine and gaming station to performing complex 3D animation and video editing, and delivering immersive Internet experiences.

In case you haven't noticed, computer technology gets more sophisticated by the month. Ten years ago, we had 100 MHz chips in computers used as servers, while today a server would demand no less than 10 times of that processing power. On the outset, there is something amiss. We have a computer that requires at least 1 GHz processor just to qualify being called a server yet a decade ago, an outfit with one-tenth the power was looked upon equally. Why this need to develop faster and more powerful hardware?

Since the birth of the PC 20 years ago, software architects have been developing applications to fully exploit the potential of computers, and at the same time, making them easier to use. As the programs that drive your PC become sophisticated and more powerful, they need an adequate infrastructure to run on. This phenomenon, whereby software drives the need for high performance hardware, is called the 'software spiral'.

The level of technology that we are at today should satisfy the needs of most users. PC owners do not need high-end machines to perform

mentioned earlier, but rather than software leading the direction of change, current and upcoming advances in the area of digital devices will mould our development efforts. Imagine using your computer to make music, not digital music, but music in the real sense with people, instruments and emotions—imagine your PC as a professional recording studio.

PCs of the future will create some terrific special-effects movies that haven't been produced yet. Post-production activities such as voice dubbing and editing will bloom into perfection. And then imagine doing all this on a PC that costs Rs 30,000.

Mobile computing is gaining strong momentum and in 2003, Baniyas (codename for our upcoming mobile processor) will make its presence felt for notebook computers worldwide. Baniyas' ability to conserve power much more efficiently than the SpeedStep technology found in mobile Pentium III processors is amazing. Very soon, once we've optimised chips to preserve more power, the potential will become limitless.

It's obvious that we have our work cut out. Though the task is challenging, I am optimistic that Intel will be very successful at meeting and accomplishing the challenges. We already have a working Pentium 4 sample operating at 3.5 GHz. If you think that's fast, you'll be further impressed once we start churning out processors running at tens of gigahertz—all of this and more within the next 10 years. ■

As narrated to Prashant Masrani. Avtar Saini, is director, South Asia, Intel Asia Electronics, Inc

